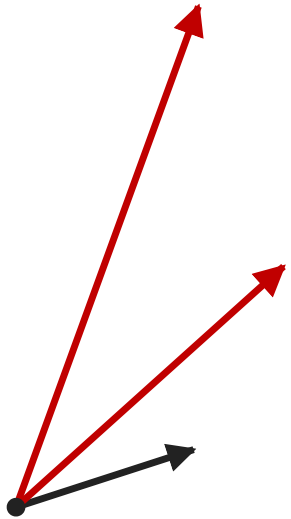
raw q, k

length grows with $\|W_q\|$

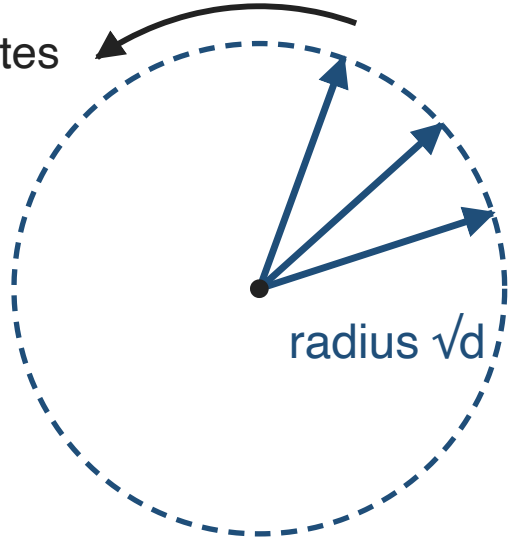


RMSNorm

per head

after QK-Norm

RoPE rotates



(unit gains)

max logit ≤ $\sqrt{d} \cdot g_q \cdot g_k$

d = 128, unit gains

max logit ≤ 11.3

g: max learned gain